EMPEL EE252 Project: Zeta Converter

A Zeta converter is a DC–DC converter that can both step up and step down voltage while keeping the output polarity the same as the input.
It provides a smooth and continuous output current, making it suitable for sensitive electronic loads.
The output voltage can be controlled by adjusting the duty cycle of the switch.
It operates with good efficiency over a wide range of input voltages.

Basic Relations of the converter:

$V_o/V_{in} = D/(1-D)$

$V_o = V_{in} \times D/(1-D)$ Vo and Vin are the output and input DC voltages respectively.

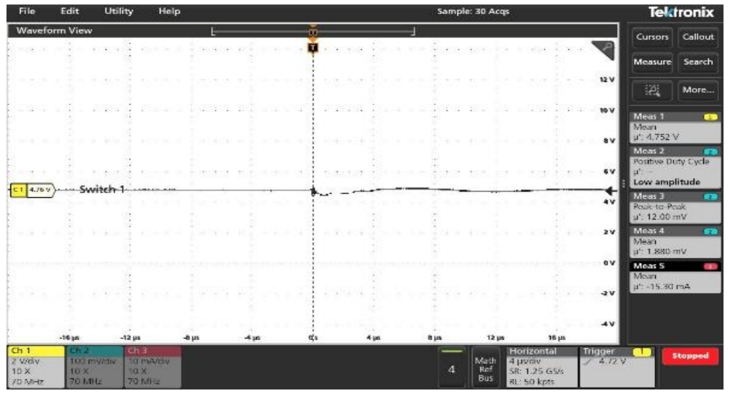
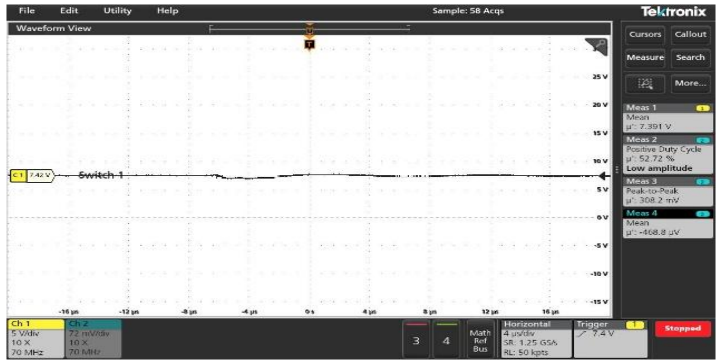
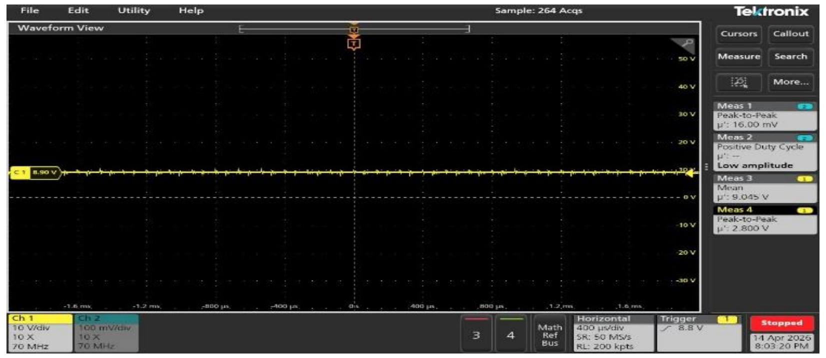
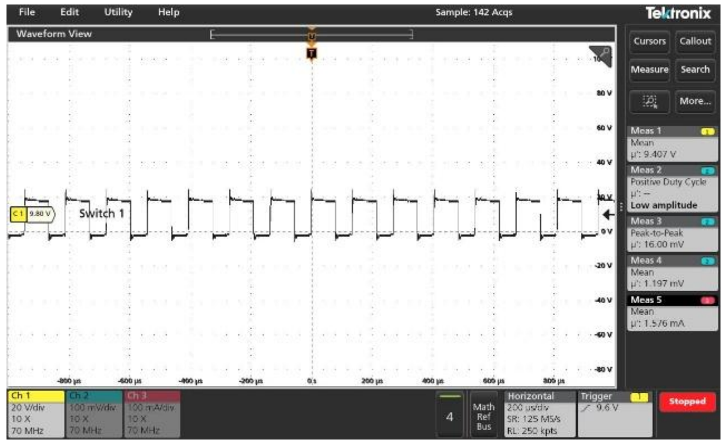
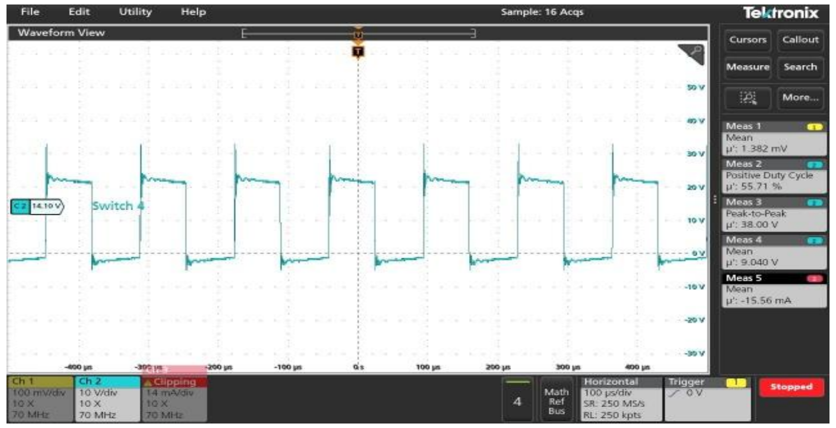
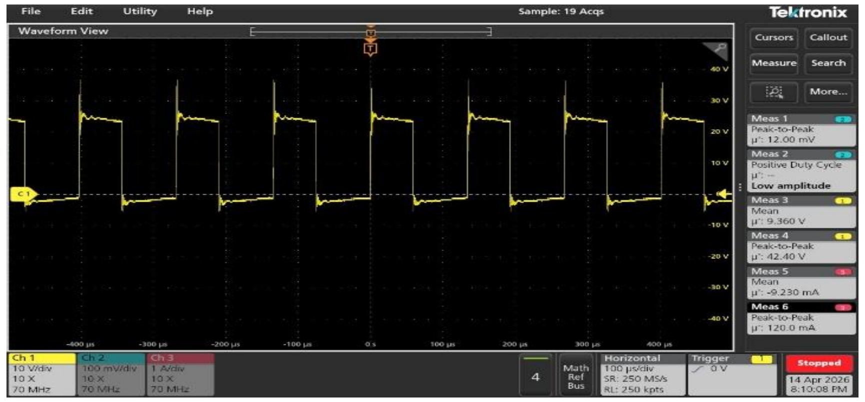
Problem Statement:

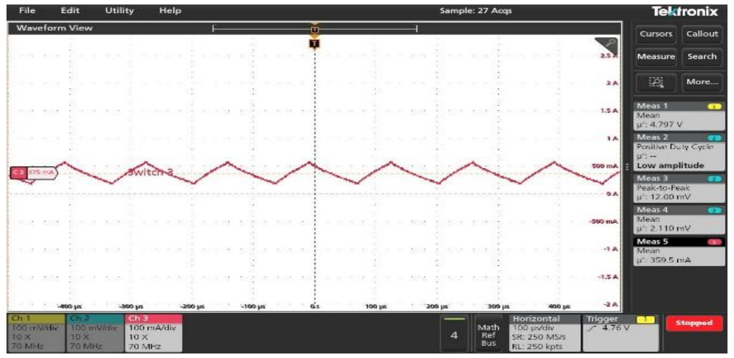
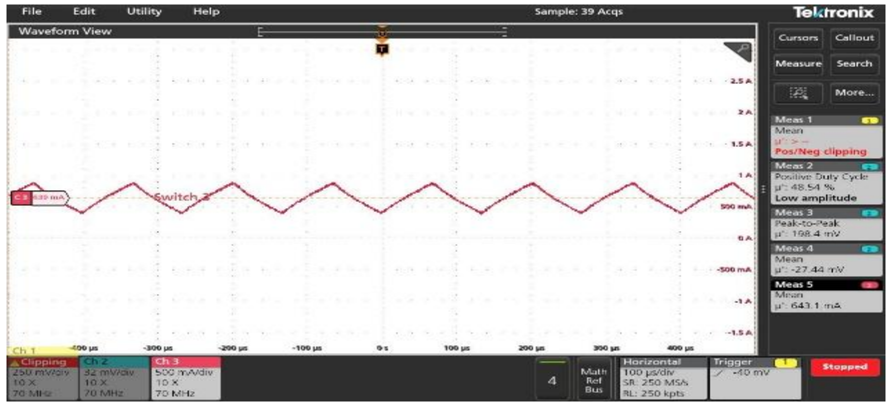
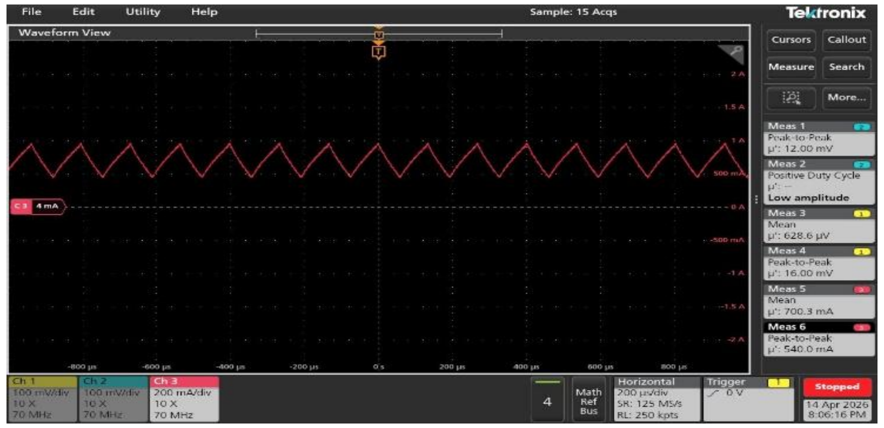
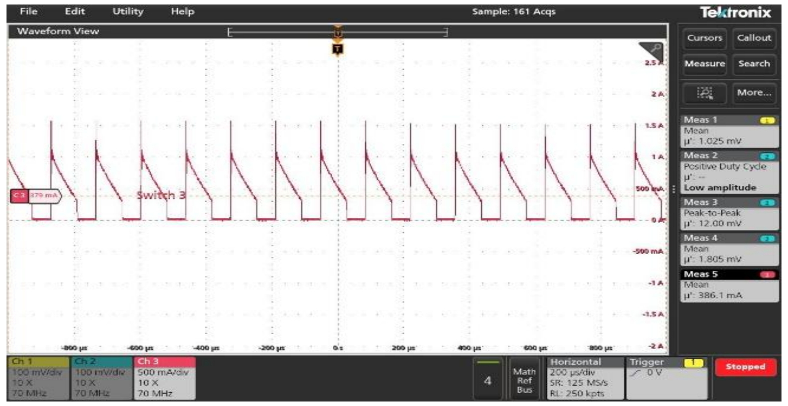
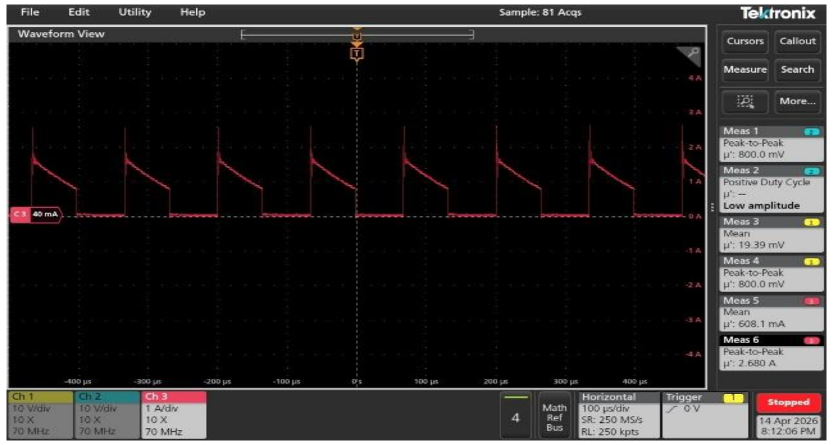
a) Specifications: Input 10 V, Output 8-15 V, Switching frequency 7.5 kHz, Output current 1 A.

Waveforms of diode current and switch voltage in CCM.

b) Reduce switching frequency to demonstrate DCM.

Part A: Given Vin= 10V, frequency = 7.5kHz

	D=40%	D=50%	D=55%
Vout	 4.76V	 7.42V	 8.90V
Vs			

IL			
Id			

Part B: Discontinuous Conduction Mode

$$\Delta I_{L1} = \frac{V_d \cdot D}{L_1 \cdot f_s};\ now\ I_o = \frac{1}{2} \cdot \Delta I_{L1} \cdot (1-D) = \frac{1}{2} \cdot \frac{V_d \cdot D}{L_1 \cdot f_s} \cdot (1-D)$$

$$\text{taking: } I_o = \frac{V_o}{R_o} \text{ and } V_o = V_d \cdot \frac{D}{1-D}$$

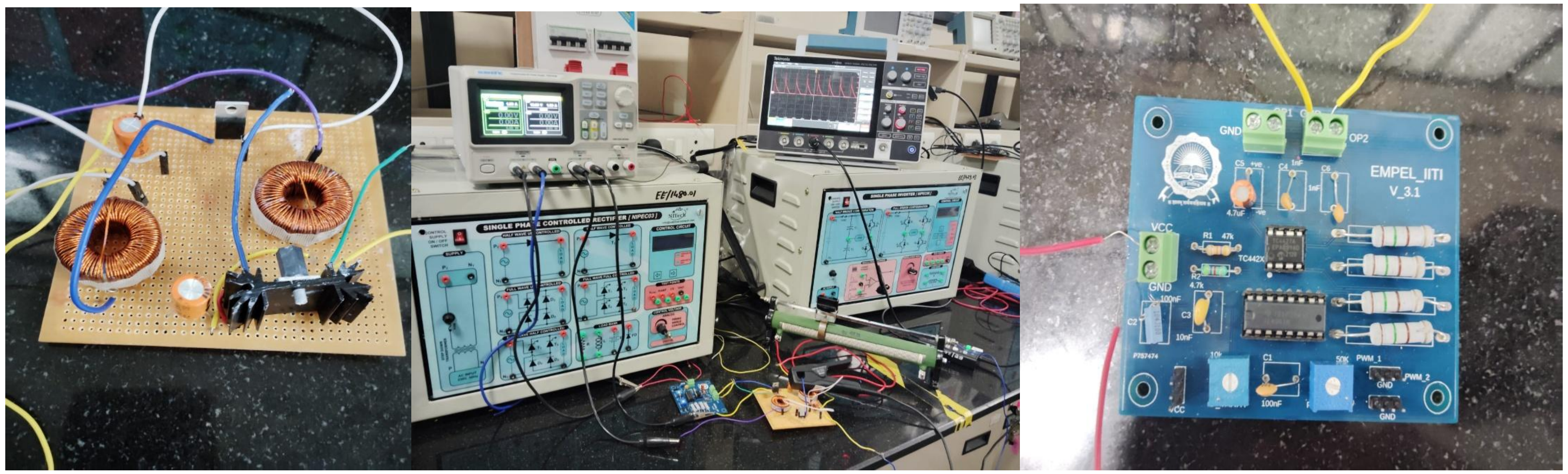
$$\frac{V_d \cdot D}{R_o(1-D)} = \frac{1}{2} \cdot \frac{V_d \cdot D}{L_1 \cdot f_s} \cdot (1-D)$$

Since in DCM ripple should be equal to the average value => $f_{\text{boundary}} = \frac{(1-D)^2 \cdot R_o}{2DL_1}$

Increasing frequency to enter DCM (discontinuous conduction mode) and noting the diode current:

FREQ=2kHz ; D=0.5

HARDWARE SETUP:



PCB used, the setup and the PWM generating module, respectively.

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